

Local structure–activity landscape roughness detects the activity-cliff failure mode that standard QSAR reliability methods miss

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Abstract

Machine-learning models of molecular potency fail on activity cliffs (pairs of similar molecules with large potency differences), and the per-compound tools meant to flag unreliable predictions miss this failure. We benchmark the standard reliability estimators (random-forest ensemble variance, distance-based applicability domain, Sheridan’s three-measure error model, and a reliability-density-neighbourhood adaptation) on 30 MoleculeACE targets and show they all rank ordinary high-error predictions but are at chance or worse at identifying activity cliffs, because each is organized around proximity to the training data, whereas cliffs sit in dense, well-sampled regions. We introduce a per-compound measure of local structure–activity landscape roughness, computed from a compound’s training neighbours without its own activity, that supplies the missing signal: it flags labelled cliffs at a median ROC-AUC of 0.69 where the prior methods score 0.35 to 0.52, stays distinct from the applicability domain under partial-correlation and mixed-effects control, and holds across three model classes, neighbourhood sizes, and distance metrics. Roughness complements rather than replaces uncertainty: a combined score keeps variance’s ranking of ordinary error and adds the cliff sensitivity it lacks. The payoff is a calibrated, cliff-aware conformal predictor: conditioning 90% prediction intervals on roughness holds coverage flat across the roughness range and restores it on cliffs, where conditioning on variance or the applicability domain leaves it below nominal. The relationship reproduces on solubility and lipophilicity, and all data and code are openly available and fully reproducible.

Contribution: We demonstrate that local landscape roughness detects the activity-cliff failure mode that ensemble variance, the applicability domain, Sheridan’s random-forest error model, and reliability-density-neighbourhood all miss, establish it as a per-compound reliability axis distinct from distance to the training data, and use it to build a conformal predictor that stays valid where QSAR models actually fail.

Keywords

activity cliffs; QSAR; structure–activity landscape; applicability domain; uncertainty quantification; conformal prediction; reliability; cheminformatics

Background

Quantitative structure–activity relationship (QSAR) modeling and, more recently, deep learning are routine tools for prioritizing compounds in drug discovery. [1,2] Almost all such models rest on the molecular similarity principle: structurally similar molecules tend to have similar properties. [3] That assumption lets a model generalize from known compounds to new ones, and it is built into the most widely used representations and algorithms, from fingerprint-based nearest-neighbor and kernel methods to message-passing graph neural networks.

Activity cliffs are the principled exception. [4,5] A cliff is a pair of structurally similar molecules whose potencies differ by orders of magnitude, a discontinuity in the structure–activity landscape that directly violates the similarity assumption. Cliffs are pervasive in medicinal-chemistry data sets and often mark the structural changes that matter most during optimization. [6] Because they break the assumption prediction relies on, cliffs are where models fail, and a recent 30-target benchmark confirmed it: machine-learning models, deep networks especially, predicted cliff compounds markedly less accurately than the rest, with no method mastering them. [7,8] The consequence is a problem of trust: a model issues a confident prediction for a compound on a cliff, exactly where it is least reliable.

The tools a modeler would use to catch unreliable predictions are organized around a single idea: a prediction is trustworthy if the query lies close to, and is well supported by, the training data. Distance-based applicability-domain (AD) analyses [14,15] measure proximity to the training set. Uncertainty estimators such as random-forest ensemble variance and conformal prediction [16,17] measure predictive disagreement. Sheridan’s widely used framework estimates a random forest’s per-compound error from three proximity/variance measures (similarity to the nearest training compound, the spread of the tree predictions, and the predicted value). [18] The reliability-density-neighbourhood method maps reliability from a query’s local neighbourhood, combining distance to the training data with the local density of training errors. [19] Data-set-level roughness indices such as ROGI [9] and its successors [10–12] report whether an entire data set is hard to model. All of these are valuable, but they share an orientation around proximity to the training data, and activity cliffs occur in dense regions among close analogs, exactly where proximity-based methods report high confidence.

Here we ask whether a per-compound quantity that does not measure proximity, the local roughness of the structure–activity landscape, can detect the cliff failure mode the standard methods miss, and whether it provides a deployable, calibrated reliability signal. By local roughness we mean how sharply activity changes in a compound’s immediate structural neighborhood: a smooth neighborhood interpolates cleanly, a rough one does not. We define a small family of functionals over a compound’s nearest training neighbors, benchmark them head-to-head against the standard reliability estimators above on the 30-target MoleculeACE data, show that roughness alone catches cliffs, that it is statistically distinct from the applicability domain, and that conditioning a conformal predictor on it keeps interval coverage valid where models fail.

Methods

Benchmark data sets

All analyses used the MoleculeACE benchmark, [7] a curated collection of bioactivity data for 30 macromolecular targets drawn from ChEMBL v29, [20] comprising 48.7K measurements (35.6K

unique structures) across 23 inhibition-constant (K_i) and 7 half-maximal effective concentration (EC_{50}) end points. For each target the regression target y is the base-10 logarithm of potency ($-\log_{10}$ of the activity in nM). Activity-cliff annotations and train/test partitions were taken unchanged from the benchmark. Following van Tilborg et al., [7] a pair of compounds was labeled an activity cliff when potencies differed by more than 10-fold ($\Delta y > 1$) and structures were similar by at least one of substructure (Tanimoto on extended-connectivity fingerprints), scaffold, or SMILES (scaled Levenshtein) similarity exceeding 0.9; a molecule is a cliff compound if it joins at least one such pair. Test sets contained 126–733 compounds per target (9,802 total); cliff prevalence ranged 9–54% (38.7% overall). For external validation, two MoleculeNet physicochemical regression benchmarks were analyzed: aqueous solubility (ESOL; 1,128 compounds [21]) and lipophilicity (4,200 compounds, log D [22]), under a deterministic Bemis–Murcko scaffold split. [23]

Molecular representations and distance metrics

Unless stated otherwise, molecules were represented by 2,048-bit extended-connectivity fingerprints of radius 2 (ECFP4 [24]) computed with RDKit, and the distance between molecules i and j was $d_{ij} = 1 - T_{ij}$ (Tanimoto), with similarity $s_{ij} = T_{ij}$. Robustness analyses additionally used ECFP6 fingerprints and a standardized 12-descriptor physicochemical vector under Euclidean distance. For every test molecule i , its k nearest neighbors $N_k(i)$ were identified among the **training** compounds; the default neighborhood size was $k = 10$.

Local landscape-roughness descriptors

Two descriptors use the query’s own activity y_i (*landscape* descriptors): a similarity-weighted local Dirichlet energy

$$R_D(i) = \frac{\sum_{j \in N_k(i)} s_{ij} (y_i - y_j)^2}{\sum_{j \in N_k(i)} s_{ij}}, \quad (1)$$

and a local Lipschitz constant $R_L(i) = \max_j |y_i - y_j| / \max(1 - s_{ij}, \varepsilon)$, $\varepsilon = 10^{-3}$. Because they require the activity being predicted, they bound what local geometry can explain rather than provide a deployable signal. Three *activity-free* descriptors use only the training neighbours’ activities and are computable before any assay of the query: the neighborhood activity dispersion $D(i) = \text{std}\{y_j : j \in N_k(i)\}$; the pairwise-SALI density $S_{\text{mean}}(i)$, the mean over neighbour pairs of $|y_a - y_b| / \max(1 - s_{ab}, \varepsilon)$ (and its max-pooled variant S_{max}) [13]; and a local Hölder regularity exponent (reported as a negative result).

Baseline reliability methods

We compared local roughness head-to-head against four established per-compound reliability estimators, all computed on the same random forest and test compounds. (i) *Ensemble variance*, the variance of the 200 tree predictions (equivalently Sheridan’s TREE_SD). (ii) The *distance-based applicability domain*, $1 - s_{\text{NN}}$, dissimilarity to the nearest training compound. (iii) *Sheridan’s three-measure error model*: [18] training compounds were binned into tertiles of nearest-training similarity, tree-prediction standard deviation, and predicted value; the out-of-bag root-mean-square error of each bin was recorded; each test compound was assigned the error of its bin. (iv) A *regression adaptation of the reliability-density neighbourhood* (RDN): [19] each training compound’s k -nearest-neighbour coverage radius was shrunk by a reliability weight $W = (1 - \text{STD}) \times \text{agreement}$ derived

from its out-of-bag prediction, and a test compound’s risk was its distance to the nearest training compound relative to that compound’s shrunk radius. RDN was developed for classification and has no published regression form, so this comparison uses our own faithful adaptation, which we flag explicitly. The data-set-level ROGI index [9] was used as a global-roughness baseline.

Predictive models and statistical analysis

A random forest (200 trees, $\sqrt{p}$ features, ECFP4; out-of-bag enabled for the baseline calibration) was the primary model; gradient-boosted trees and a radial-basis-function support-vector regressor were used for the model-class generality analysis. A Graph Isomorphism Network (GIN) provided a graph baseline. All models used a fixed random seed; hyperparameters were fixed a priori rather than tuned, since the analysis compares the relative structure of per-compound error rather than absolute accuracy. Per-compound out-of-sample error was $e_i = |\hat{y}_i - y_i|$. For each reliability score and target we computed the ROC-AUC for ranking top-quartile-error compounds and for flagging labelled cliffs, aggregated as the median across 30 targets. Spearman correlations between descriptors and per-compound error were aggregated as the median with a Wilcoxon signed-rank test; partial Spearman correlations and a linear mixed-effects model controlled for the applicability domain, with effect sizes reported as 95% confidence intervals (exact p values are uninformative at $n = 9,802$). All analyses were repeated for $k \in \{5, 10, 15, 20, 30\}$ and three distance metrics. Software versions and a single `reproduce.sh` entry point are given under Availability of data and materials.

Results and Discussion

Activity cliffs concentrate prediction error

Across the 30 benchmarks the random forest predicted cliff compounds less accurately than non-cliff compounds (mean absolute error 0.62 vs 0.46 log units; cliff error higher on 29 of 30 targets; Wilcoxon $p = 3.7 \times 10^{-9}$). Cliffs are a generic, reproducible failure mode; the question is whether it can be anticipated, and whether existing reliability tools anticipate it.

Roughness detects the cliff failure that prior reliability methods miss

We benchmarked local roughness head-to-head against the four standard per-compound reliability estimators on two tasks: ranking the highest-error predictions in general, and flagging labelled activity cliffs (Table 1). On ordinary error every proximity- or variance-based method performs well, with ensemble variance strongest (median ROC-AUC 0.72); roughness is the weakest of the set (0.60). On activity cliffs the picture inverts completely. Ensemble variance and Sheridan’s full three-measure error model are at chance (0.51), the distance-based applicability domain is actively inverted (0.35), and the reliability-density-neighbourhood adaptation is below chance (0.46), because all four are organized around proximity to the training data and cliffs sit among close, densely sampled analogs. Local roughness is the only estimator that detects cliffs (0.69). A combined score (within-target standardized variance plus roughness) keeps variance’s ranking of ordinary error (0.70) and adds the cliff sensitivity variance lacks (0.58). Roughness is therefore not a better general-error estimator; its distinct, demonstrable value is that it captures the cliff failure mode that the standard reliability methods systematically miss or invert.

Table 1: Head-to-head reliability benchmark. Median per-target ROC-AUC (30 targets) for ranking top-quartile-error compounds and for flagging labelled activity cliffs. Every prior method ranks ordinary error but is at chance or worse on cliffs; local roughness alone detects cliffs, and the combined score inherits both.

Per-compound reliability method	Top-quartile error	Activity cliffs
Ensemble variance (Sheridan TREE_SD) [18]	0.72	0.51
Sheridan three-measure error model [18]	0.64	0.51
Distance-based applicability domain [15]	0.62	0.35
Reliability-density neighbourhood (regression adaptation) [19]	0.62	0.46
Local roughness (this work)	0.60	0.69
Combined (variance + roughness)	0.70	0.58

Activity-free roughness predicts per-compound error

Treating each target as an independent replicate, the activity-free neighborhood dispersion D was positively correlated with per-compound random-forest error on all 30 targets (median Spearman $\rho = 0.28$, FDR-adjusted $p = 2.7 \times 10^{-9}$), and the pairwise-SALI density behaved likewise (median $\rho = 0.16$), each exceeding the applicability-domain baselines ($\rho \approx 0.18$ – 0.19) (Figure 1A). The landscape descriptors that use the query’s own activity correlate far more strongly (Dirichlet energy median $\rho = 0.65$) but, requiring the potency being predicted, bound rather than provide the deployable signal. That a purely structural quantity tracks error on every target shows cliff-prone regions announce themselves before any potency is measured.

Roughness is distinct from the applicability domain

The dispersion result could reflect data sparsity, so we tested whether roughness predicts error after the applicability domain is controlled (Figure 1B). Partial Spearman correlations removing nearest-neighbor similarity and local density left the signal intact: the Dirichlet energy retained essentially all of its association ($0.65 \rightarrow 0.64$), neighborhood dispersion fell only modestly ($0.28 \rightarrow 0.23$), and the pairwise-SALI density rose ($0.16 \rightarrow 0.23$). A linear mixed-effects model agreed under simultaneous control of nearest-neighbor similarity, local density, molecular size, and a per-target random intercept: standardized neighborhood dispersion carried a coefficient of $+0.26$ (95% CI $[0.24, 0.28]$), the pairwise-SALI density $+0.13$ ($[0.11, 0.15]$), and the landscape Dirichlet energy $+0.73$ ($[0.72, 0.75]$), all $p < 0.001$. Local roughness is thus a distinct axis of predictability, not a re-expression of distance to the training set. For cliffs the two notions are opposed: nearest-neighbor similarity, used in its standard extrapolation orientation, flags cliffs at an AUC of 0.35, because cliffs concentrate among the close analogs the applicability domain deems safe (Figure 1C).

Generality across model classes

Repeating the analysis with gradient-boosted trees and a support-vector regressor on the same features gave the same picture (Table 2): the landscape Dirichlet energy correlated with per-compound error at median ρ of 0.65, 0.58, and 0.55; the activity-free dispersion at 0.28, 0.25, and 0.24; and the applicability-domain-controlled partial correlation at 0.23, 0.21, and 0.18. The signal is consistent across model classes rather than an idiosyncrasy of a single learner.

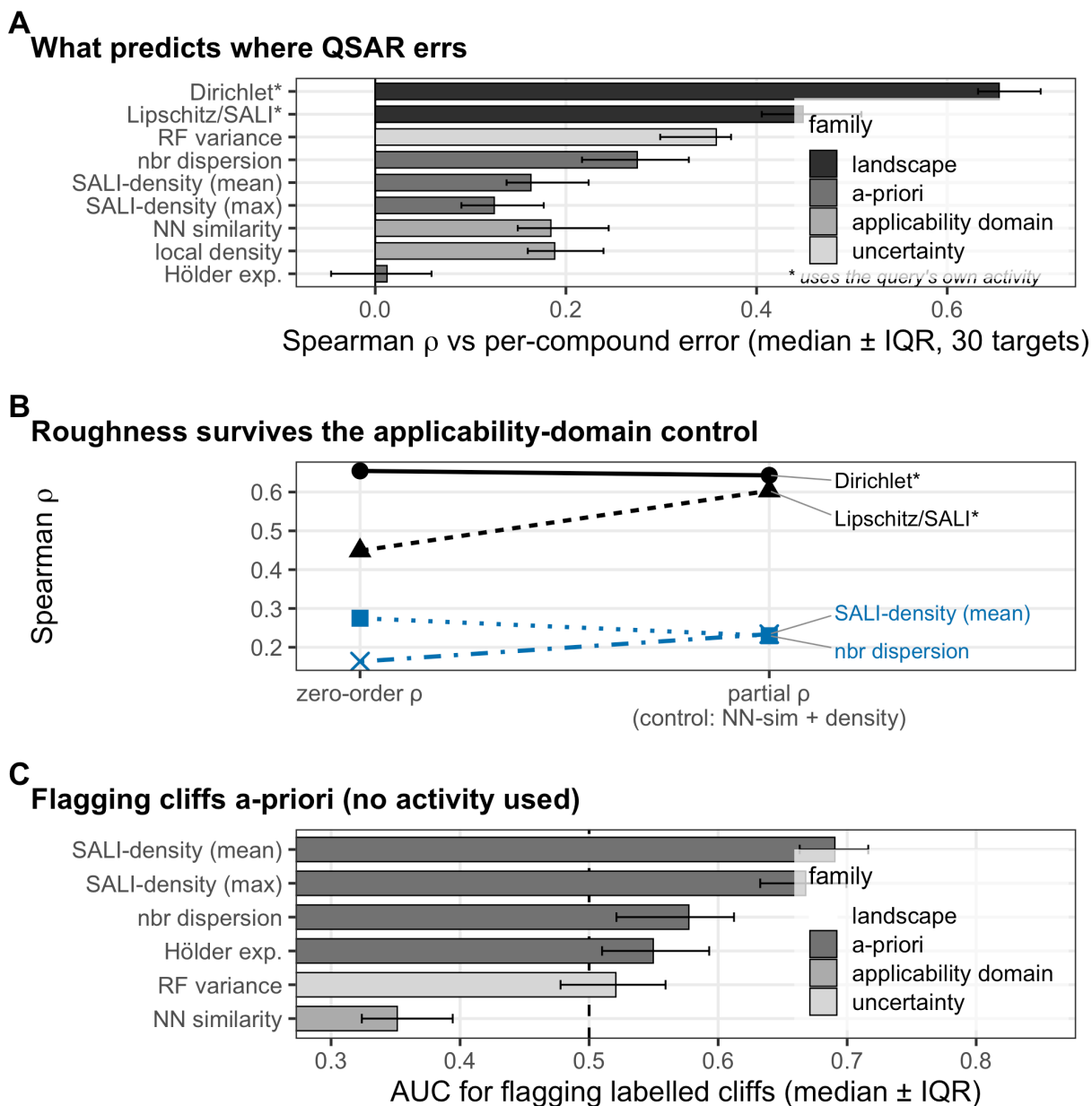


Figure 1: Local structure–activity landscape roughness predicts per-compound QSAR error across 30 bioactivity targets. (A) Median Spearman correlation between each descriptor and per-compound random-forest error (median $\pm$ interquartile range); landscape descriptors use the query’s own activity, activity-free descriptors do not. (B) Zero-order versus applicability-domain-controlled partial correlations; the relationship is retained or strengthened. (C) ROC-AUC for flagging labelled cliffs from activity-free descriptors; nearest-neighbor similarity falls below 0.5 in the extrapolation orientation because cliffs cluster among close analogs.

Table 2: Median Spearman correlation (30 targets) between local roughness and per-compound error for three model classes. The final column is the neighborhood-dispersion correlation after controlling for the applicability domain.

Model	Dirichlet (landscape)	Nbr. dispersion (activity-free)	Nbr. dispersion AD
Random forest	0.65	0.28	0.23
Gradient boosting	0.58	0.25	0.21
SVR (RBF)	0.55	0.24	0.18

Operational triage

The head-to-head translates into a concrete triage benefit (Figure 2). To recover the highest-error predictions in general, ensemble uncertainty is the strongest single flag (top 20% recovers 38% of top-quartile errors, a 1.9-fold enrichment), with roughness (33%) and the applicability domain (32%) close behind. For activity cliffs the picture inverts: ranking the top 20% by roughness recovers 33% of labelled cliffs (1.6-fold), where ensemble uncertainty performs at chance (1.0-fold) and the applicability domain below it (0.45-fold).

Operational triage: a structure-only roughness flag captures more problem compounds than applicability domain or uncertainty

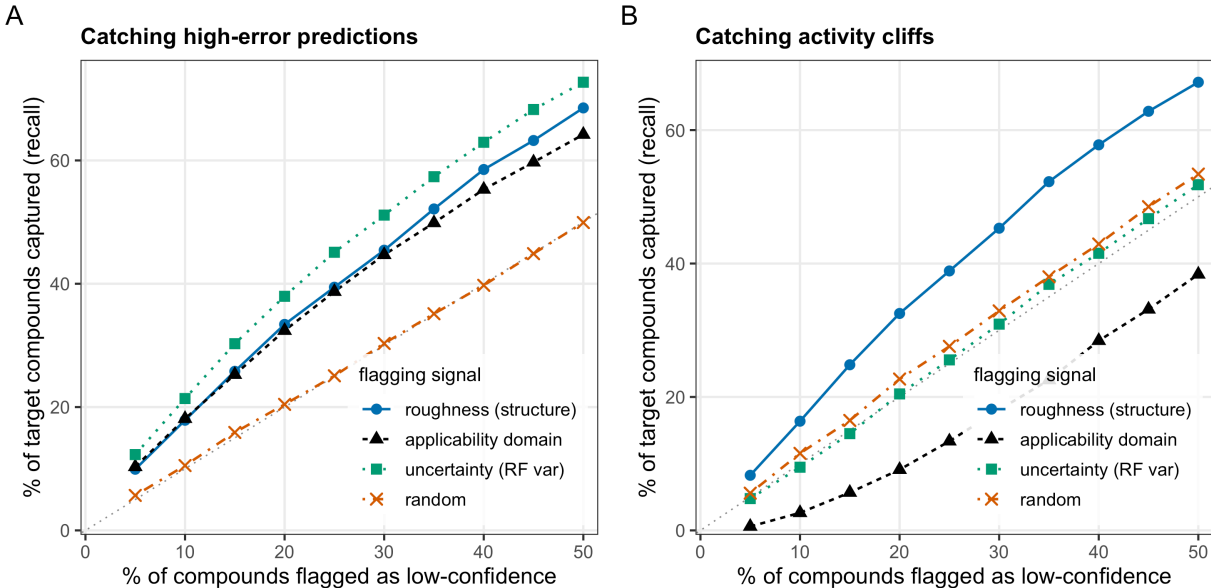


Figure 2: Operational triage value of an activity-free roughness flag. Fraction of (A) high-error predictions and (B) labelled activity cliffs recovered as a function of the percentage of compounds flagged as low-confidence, averaged over 30 targets, ranking by roughness, the applicability domain, random-forest tree variance, or at random.

Robustness and external validation

Across five neighborhood sizes ($k = 5-30$) and three distance metrics, the relationship was stable, and the applicability-domain-controlled partial correlation remained positive in all fifteen combinations (Figure 3). Applied unchanged to the two MoleculeNet physicochemical benchmarks

under scaffold splits (Figure 4), the central result reproduced: landscape roughness predicted per-compound error on solubility and lipophilicity (partial $\rho = 0.60$ and 0.64), matching the bioactivity value. The activity-free dispersion transferred more modestly (partial $\rho = 0.19$ on solubility, 0.07 on lipophilicity), consistent with the modest, target-dependent magnitude of the activity-free signal throughout.

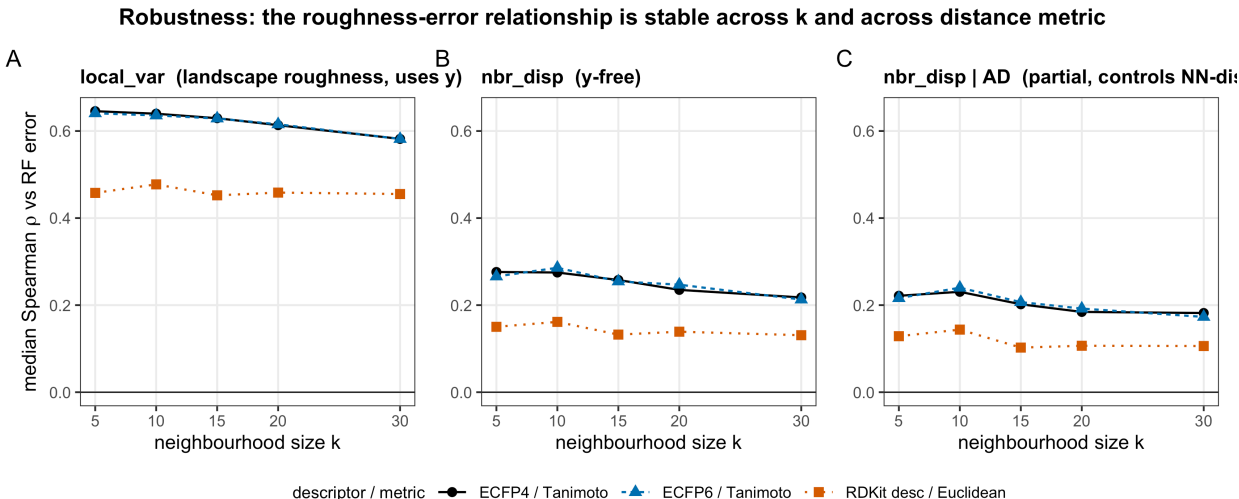


Figure 3: Stability of the roughness-error relationship across neighborhood size k and molecular distance metric (ECFP4/Tanimoto, ECFP6/Tanimoto, and a 12-descriptor Euclidean representation), for a landscape roughness measure, the activity-free dispersion, and the dispersion after controlling for the applicability domain.

Roughness keeps prediction intervals valid on activity cliffs

The head-to-head advantage becomes decisive for calibrated prediction intervals. A conformal predictor turns a point estimate into an interval with a coverage guarantee, but the guarantee is only marginal, and the per-compound variable chosen to make it conditional decides where the coverage lands. Conditioning on tree variance, the obvious choice, does not solve the problem: a Mondrian predictor stratified by variance left coverage falling with local roughness and its cliff coverage at 86.0%, no better than the 87.2% of unconditional intervals (Figure 5). The applicability domain is worse, dropping cliff coverage to 82.0% as a locally-weighted scale. Only conditioning on the activity-free roughness fixes it: a roughness-stratified Mondrian predictor held coverage flat across the whole roughness range, lifted cliff coverage to 89.5% (improving 27 of 30 targets) at a 6% median increase in width, and a locally-weighted roughness score reached 91.2%. Conditioning on the combined variance-plus-roughness score keeps coverage flat at no width cost while recovering most of the cliff coverage. The contribution is not the conformal machinery, which is standard, but the identification of the per-compound scale: where the landscape is rough, local roughness makes coverage valid and the conventional proximity-based signals do not.

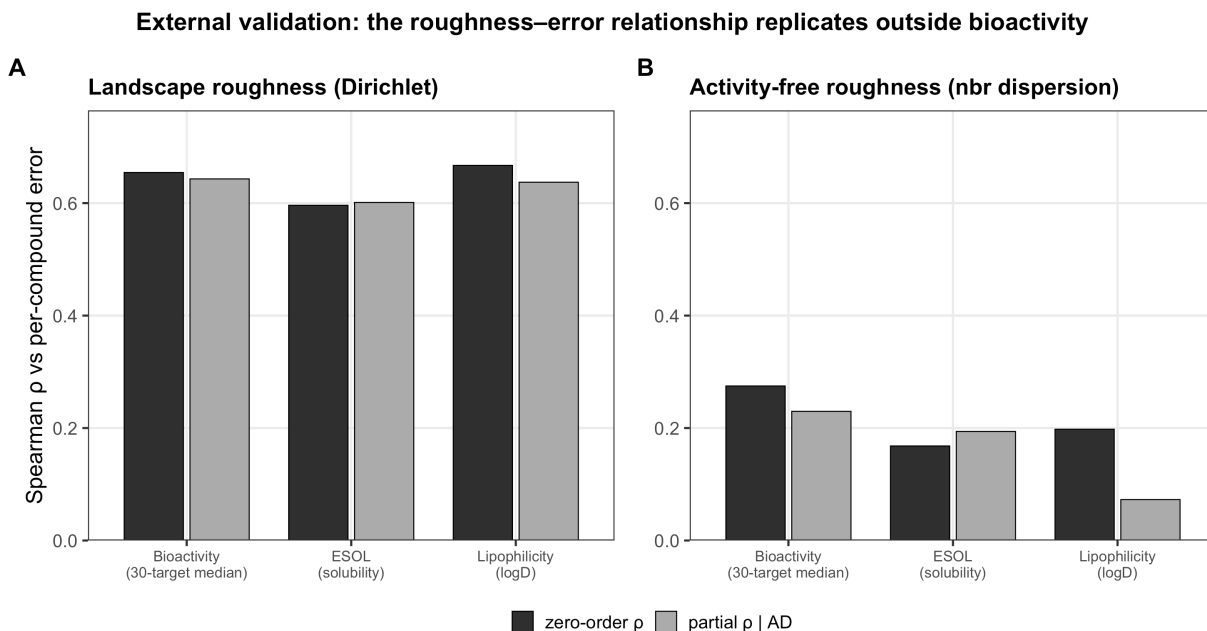


Figure 4: External validation on physicochemical regression benchmarks. Zero-order and applicability-domain-controlled (partial) Spearman correlations between roughness and per-compound error for (A) the landscape Dirichlet energy and (B) the activity-free dispersion, for the 30-target bioactivity median and for aqueous solubility and lipophilicity.

Roughness, not tree variance or the applicability domain, is the per-compound scale that keeps coverage valid on cliffs

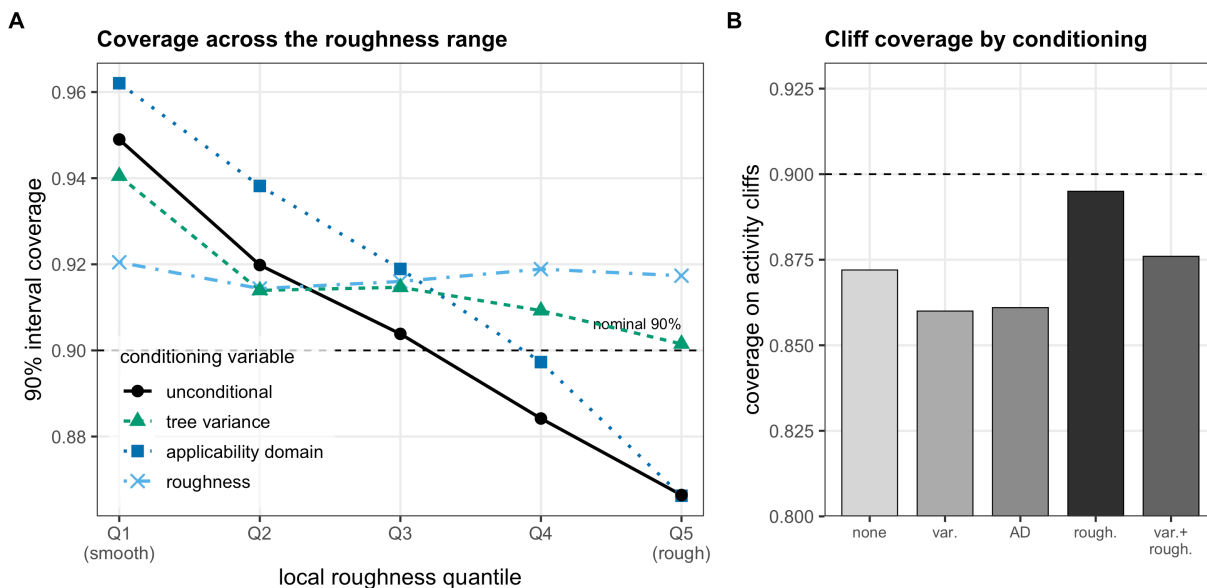


Figure 5: Roughness is the per-compound scale for cliff-valid conformal intervals. (A) Coverage of 90% inductive-conformal intervals versus local-roughness quantile, for unconditional intervals and Mondrian intervals conditioned on tree variance, the applicability domain, or roughness. (B) Coverage on labelled activity cliffs for the same conditionings and the combined score. Median over 30 targets and 50 calibration/evaluation splits.

Comparison with a graph-convolutional model

Consistent with prior benchmarking, a graph isomorphism network gave higher mean absolute error than the random forest on all 30 targets (0.71 versus 0.52 log units). Resolving the gap by roughness, the network’s disadvantage was largest in the smoothest quartile (excess 0.25 log units) and smallest in the roughest (0.13), and its deficit on cliffs (0.16) was smaller than on non-cliff compounds (0.21). A tuned regimen did not systematically help (mean MAE 0.72; better on 14 of 30 targets). The familiar advantage of descriptors over graph networks on these data is thus concentrated in smooth, well-sampled regions, not in superior treatment of cliffs, where both classes fail comparably. Neither network was architecture-optimized, so this is offered as a supporting observation (Figure 6).

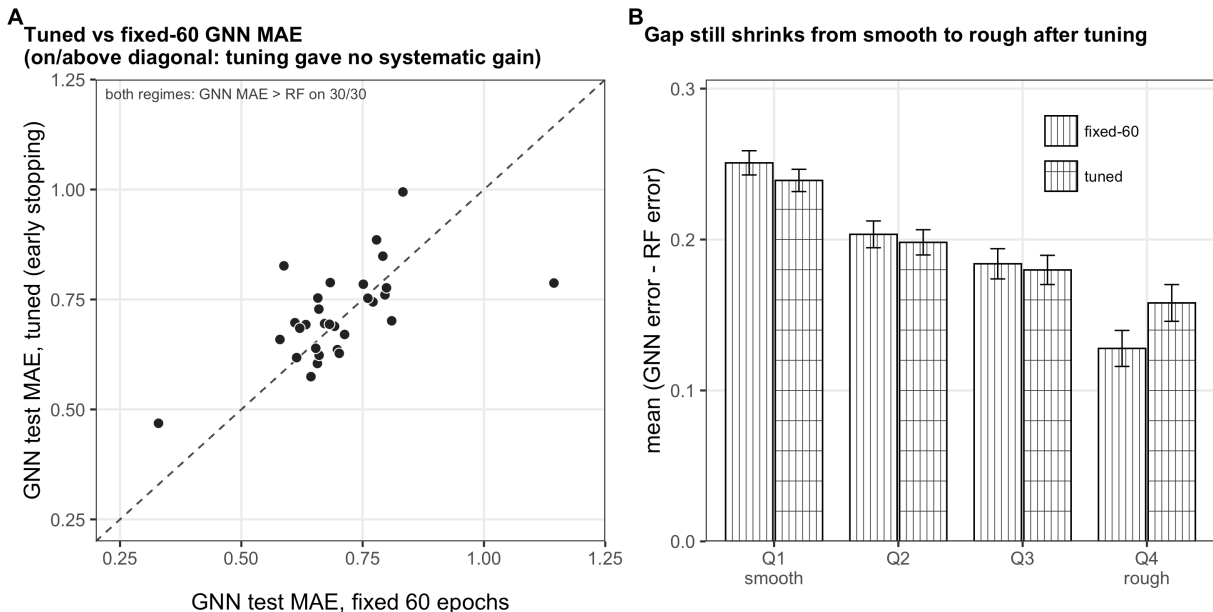


Figure 6: Comparison with a graph-convolutional baseline. (A) Per-target test mean absolute error of the graph isomorphism network under a fixed schedule versus validated early stopping; both exceed the random forest on all 30 targets. (B) Mean (network – random forest) per-compound error by within-target roughness quartile; the deficit is largest where the landscape is smooth.

Local versus global roughness

A data-set-level description of roughness does not substitute for the per-compound one. The global ROGI index correlated with the cliff fraction across the 30 benchmarks (Spearman $\rho = 0.60$) but not with mean random-forest error ($\rho = -0.12$, $p = 0.54$): a single scalar cannot resolve which compounds within a set will be mispredicted.

Limitations

The activity-free effect sizes, although consistent across targets, are modest, and the study is confined to regression on curated benchmarks. The reliability-density-neighbourhood comparison uses

our regression adaptation of a method developed for classification. The graph-network comparison is a supporting observation only, and one construct (the local Hölder exponent) did not predict error (median $\rho = 0.01$).

Conclusions

Local landscape roughness, computed from a compound’s training neighbours without its own activity, detects the activity-cliff failure mode that the standard per-compound reliability methods, ensemble variance, the applicability domain, Sheridan’s random-forest error model, and reliability-density-neighbourhood, are all at chance or worse at identifying. It is a distinct reliability axis, complementary to the proximity-based signals in use, and conditioning a conformal predictor on it keeps prediction-interval coverage valid on the cliffs those signals miss. Because it needs no assay of the query and is reproducible from open data and code, it is a practical addition to the QSAR reliability toolkit precisely where models are most often, and most confidently, wrong.

List of abbreviations

QSAR: quantitative structure–activity relationship; AD: applicability domain; RF: random forest; GIN: graph isomorphism network; SALI: structure–activity landscape index; RDN: reliability-density neighbourhood; ROGI: roughness index; ECFP: extended-connectivity fingerprint; AUC: area under the receiver-operating-characteristic curve; MAE: mean absolute error; OOB: out-of-bag; CI: confidence interval.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The bioactivity benchmark (30 ChEMBL targets with activity-cliff annotations and fixed train/test splits) is the publicly available MoleculeACE data set, [7] and the external benchmarks (ESOL, Lipophilicity) are distributed with MoleculeNet. [21,22] All analysis code, the derived per-compound descriptor and error tables, and scripts that regenerate every figure and table are openly available under the MIT licence at <https://github.com/krishnatheaverage/qsar-landscape-roughness>, with a single `reproduce.sh` entry point and a pinned environment (Python 3.12, RDKit, scikit-learn, PyTorch, statsmodels). A snapshot is archived on Zenodo (concept DOI 10.5281/zenodo.20603136, which resolves to the latest version).

Competing interests

The author declares no competing interests.

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Authors’ contributions

K.H. conceived the study, implemented the analyses, and wrote the manuscript. The author read and approved the final manuscript.

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